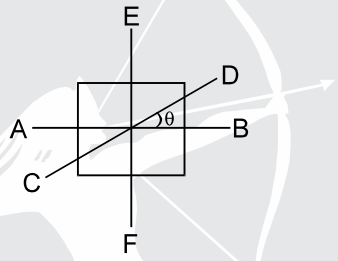
SYLLABUS : ROTATIONAL BODY DYNAMICS

1. The moment of inertia of a hollow cubical box of mass M and side a about an axis passing through the centres of two opposite faces is equal to

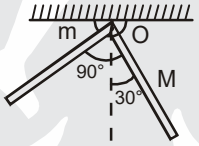
(A) $\frac{5Ma^2}{3}$ (B) $\frac{5Ma^2}{6}$ (C) $\frac{5Ma^2}{12}$ (D) $\frac{5Ma^2}{18}$

2. Let I be the moment of inertia of a uniform square plate about an axis AB that passes through its centre and is parallel to two of its sides. CD is a line in the plane of the plate that passes through the centre of the plate and makes an angle θ with AB as shown in figure. The moment of inertia of the plate about the axis CD is then equal to



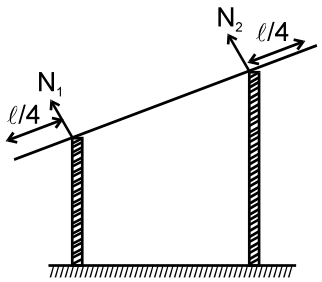
(A) I (B) $I \sin^2\theta$ (C) $I \cos^2\theta$ (D) $I \cos^2(\theta/2)$

3. Two uniform rods of equal length but different masses are rigidly joined to form an L–shaped body, which is then pivoted about O as shown in the figure. If in equilibrium the body is in the shown configuration, ratio M/m will be:



(A) 2 (B) 3 (C) $\sqrt{2}$ (D) $\sqrt{3}$

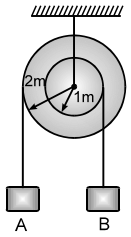
4. A uniform rod of length ℓ is placed symmetrically on two walls as shown in figure. The rod is in equilibrium. If N_1 and N_2 are the normal forces exerted by the walls on the rod then



(A) $N_1 > N_2$ (B) $N_1 > N_2$

(C) $N_1 = N_2$ (D) N_1 and N_2 would be in the vertical directions.

5. In the pulley system shown in the figure, if radii of the bigger and smaller pulley are 2 m and 1 m respectively and the acceleration of block A is 5 m/s^2 in the downward direction, then the acceleration of block B will be:



(A) 0 m/s^2 (B) 5 m/s^2 (C) 10 m/s^2 (D) 5/2 m/s^2

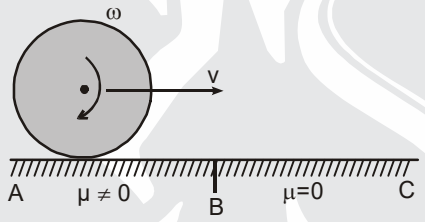
6. The angular momentum of a particle about origin is varying as L = 4t + 8 (SI units) when it moves along a straight line y = x – 4 (x,y in meters). The magnitude of force acting on the particle would be :

(A) 1 N (B) 2 N (C) $\sqrt{2}$ N (D) $\sqrt{3}$ N

7. When a person throws a meter stick it is found that the centre of the stick is moving with a speed of 10 m/s vertically upwards & left end of stick with a speed of 20 m/s vertically upwards. Then the angular speed of the stick is:

(A) 20 rad/ sec (B) 10 rad/sec (C) 30 rad/sec (D) none of these

8. As shown in the figure, a uniform disc of mass m is rolling without slipping with a angular velocity ω. The portion AB is rough and BC is smooth. When it crosses point B disc will be in :

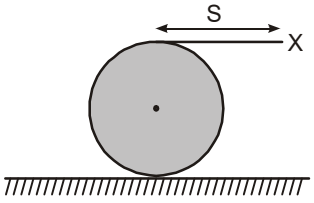


(A) translational motion only (B) pure rolling motion

(C) rotational motion only (D) none of these

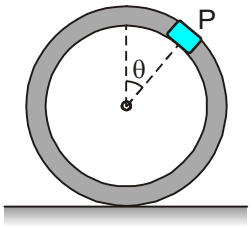
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9. A large spool of rope lies on the ground as shown in the fig. The end, labelled X is pulled a distance S in the horizontal direction. The spool rolls without slipping. The centre of mass of the spool moves a distance



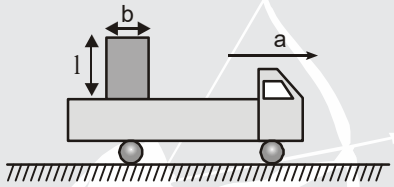
(A) 2 S (B) S (C) $\frac{S}{2}$ (D) $\frac{S}{4}$

10. A small block of mass 'm' is rigidly attached at 'P' to a ring of mass '3m' and radius 'r'. The system is released from rest at $\theta = 90°$ and rolls without sliding. The angular acceleration of ring just after release is



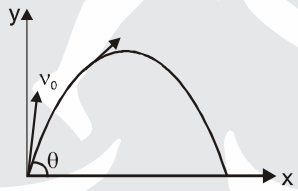
(A) $\frac{g}{4r}$ (B) $\frac{g}{8r}$ (C) $\frac{g}{3r}$ (D) $\frac{g}{2r}$

11. A box of dimensions ℓ and b is kept on a truck moving with an acceleration a. If box does not slide, maximum acceleration for it to remain in equilibrium (w.r.t.truck) is :



(A) $\frac{g\,\ell}{b}$ (B) $\frac{g\,b}{\ell}$ (C) g (D) none of these

12. A small particle of mass m is projected at an angle θ with the x-axis with an initial velocity v_0 in the x-y plane as shown in the figure. At a time $t < \frac{v_0 \sin\theta}{g}$, the angular momentum of the particle is



(A) $-mg\, v_0\, t^2 \cos\theta\, \hat{j}$ (B) $mg\, v_0\, t \cos\theta\, \hat{k}$

(C) $-\frac{1}{2} mg\, v_0\, t^2 \cos\theta\, \hat{k}$ (D) $\frac{1}{2} mg\, v_0\, t^2 \cos\theta\, \hat{i}$

where $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along x, y and z-axis respectively.

13. A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect is at rest at a point near the rim of the disc. The insect now moves along a diameter of the disc to reach its other end. During the journey of the insect, the angular speed of the disc:

(A) remains unchanged (B) continuously decreases

(C) continuously increases (D) first increases and then decreases

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14. A mass m hangs with the help of a string wrapped around a pulley on a frictionless bearing. The pulley has mass m and radius R. Assuming pulley to be a perfect uniform circular disc, the acceleration of the mass m, if the string does not slip on the pulley, is :

(A) $\frac{3}{2}g$ (B) g (C) $\frac{2}{3}g$ (D) $\frac{g}{3}$

15. A pulley of radius 2m is rotated about its axis by a force $F = (20t - 5t^2)$ newton (where t is measured in seconds) applied tangentially. If the moment of inertia of the pulley about its axis of rotation is 10 kg m^2, the number of rotations made by the pulley before its direction of motion is reversed, is :

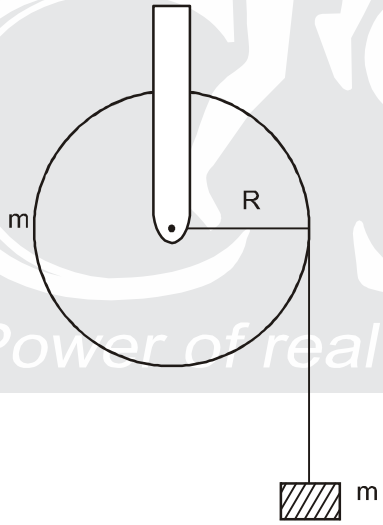
(A) less than 3 (B) more than 3 but less than 6

(C) more than 6 but less than 9 (D) more than 9

16. A particle of mass 'm' is projected with a velocity υ making an angle of 30º with the horizontal. The magnitude of angular momentum of the projectile about the point of projection when the particle is at its maximum height 'h' is :

(A) zero (B) $\frac{m\upsilon^3}{\sqrt{2}g}$ (C) $\frac{\sqrt{3}}{16}\frac{m\upsilon^3}{g}$ (D) $\frac{\sqrt{3}}{2}\frac{m\upsilon^2}{g}$

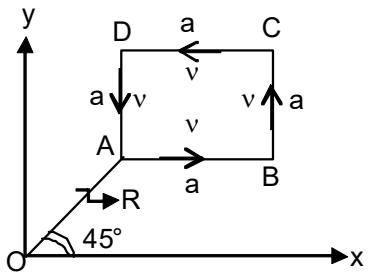
17. A mass 'm' supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release ?



(A) $\frac{2g}{3}$ (B) $\frac{g}{2}$ (C) $\frac{5g}{6}$ (D) g

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18. A particle of mass m is moving along the side of square of side 'a' with a uniform speed v in the x-y plane as shown in the figure :



Which of the following statements is false for the angular momentum $\vec{L}$ about the origin ?

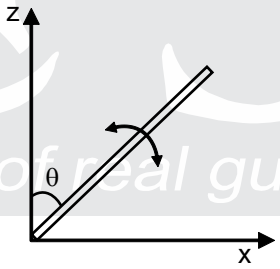
(A) $\vec{L} = mv\left[\frac{R}{\sqrt{2}} - a\right]\hat{k}$ when the particle is moving from C to D.

(B) $\vec{L} = mv\left[\frac{R}{\sqrt{2}} + a\right]\hat{k}$ when the particle is moving from B to C.

(C) $\vec{L} = \frac{mv}{\sqrt{2}}R\ \hat{k}$ when the particle is moving from D to A.

(D) $\vec{L} = -\frac{mv}{\sqrt{2}}R\ \hat{k}$ when the particle is moving from A to B.

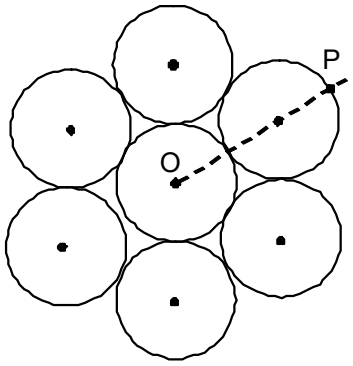
19. A slender uniform rod of mass M and length *l* is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. the angular acceleration of the rod when it makes an angle θ with the vertical is.



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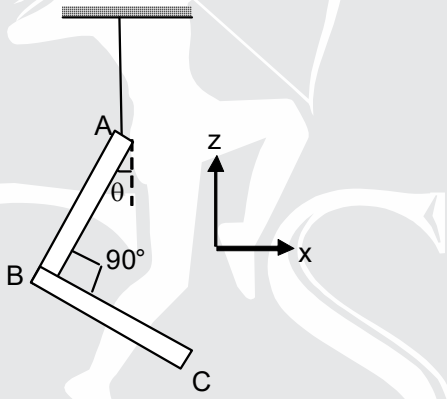
(A) $\frac{2g}{3l}\cos\theta$ (B) $\frac{3g}{2l}\sin\theta$ (C) $\frac{2g}{3l}\sin\theta$ (D) $\frac{3g}{2l}\cos\theta$

20. Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is :



(A) $\frac{73}{2}MR^2$ (B) $\frac{181}{2}MR^2$ (C) $\frac{19}{2}MR^2$ (D) $\frac{55}{2}MR^2$

21. An L-shaped object, made of thin rods of uniform mass density, is suspended with a string as shown in figure. If AB = BC, and the angle made by AB with downward vertical is θ, then :



(A) $\tan\theta = \frac{1}{2}$ (B) $\tan\theta = \frac{1}{3}$ (C) $\tan\theta = \frac{1}{2\sqrt{3}}$ (D) $\tan\theta = \frac{2}{\sqrt{3}}$

22. A homogeneous solid cylindrical roller of radius R and mass M is pulled on a cricket pitch by a horizontal force. Assuming rolling without slipping, angular acceleration of the cylinder is :

(A) $\frac{F}{3mR}$ (B) $\frac{F}{2mR}$ (C) $\frac{3F}{2mR}$ (D) $\frac{2F}{3mR}$

23. The linear mass density of a thin rod AB of length L varies from A to B as $\lambda(x) = \lambda_0\left(1+\frac{x}{L}\right)$, where x is the distance from A. If M is the mass of the rod then its moment of inertia about an axis passing through A and perpendicular to the rod is

(A) $\frac{2}{5}ML^2$ (B) $\frac{5}{12}ML^2$ (C) $\frac{3}{7}ML^2$ (D) $\frac{7}{18}ML^2$

24. Consider two uniform discs of the same thickness and different radii $R_1 = R$ and $R_2 = \alpha R$ made of the same material. If the ratio of their moments of inertia I_1 and I_2, respectively, about their axes is $I_1 : I_2 = 1 : 16$ then the value of α is

(A) 2 (B) 4 (C) $2\sqrt{2}$ (D) $\sqrt{2}$

25. Moment of inertia (M.I.) of four bodies, having same mass and radius, are reported as ;

I_1 = M.I. of thin circular ring about its diameter.

I_2 = M.I. of circular disc about an axis

perpendicular to the disc and going through the centre,

I_3 = M.I. of solid cylinder about its axis and

I_4 = M.I. of solid sphere about its diameter.

Then :

(A) $I_1 + I_3 < I_2 + I_4$ (B) $I_1 + I_2 = I_3 + \frac{5}{2}I_4$ (C) $I_1 = I_2 = I_3 > I_4$ (D) $I_1 = I_2 = I_3 < I_4$

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ANSWER KEY

1.	(D)	**2.**	(A)	**3.**	(D)	**4.**	(C)	**5.**	(D)
6.	(C)	**7.**	(A)	**8.**	(B)	**9.**	(C)	**10.**	(B)
11.	(B)	**12.**	(C)	**13.**	(D)	**14.**	(C)	**15.**	(B)
16.	(C)	**17.**	(B)	**18.**	(A,C)	**19.**	(B)	**20.**	(B)
21.	(B)	**22.**	(D)	**23.**	(D)	**24.**	(A)	**25.**	(C)